UNIT I — NATURAL LANGUAGE PROCESSING (NLP)

Detailed Easy-English Exam Notes

Syllabus Covered

Introduction to Natural Language Processing, NLP applications, challenges in NLP, NLP pipeline, text preprocessing, tokenization, stop-word removal, stemming, and lemmatization.

1. Introduction to Natural Language Processing

Natural language means the language humans normally use to communicate, such as English, Hindi, Urdu, Arabic, French, and Chinese.

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that helps computers understand, process, analyze, and generate human language in the form of text or speech.

> **Easy definition:** NLP is the technology that allows computers to understand and work with human language.

Why do we need NLP?

- A huge amount of information is available as text or speech.
- Humans cannot manually read and analyze millions of documents efficiently.
- NLP allows computers to process large amounts of language automatically.
- It helps systems find meaning, opinions, entities, topics, and useful information from text.

Example

Suppose a company receives one million customer reviews. Without NLP, people would need to read the reviews manually. With NLP, a system can automatically classify reviews as positive, negative, or neutral.

2. NLP and Artificial Intelligence

NLP is an important area of Artificial Intelligence. NLP systems can use rule-based methods, Machine Learning, Deep Learning, neural networks, transformers, and Large Language Models depending on the task.

3. Applications of NLP

Machine Translation: Automatically converts text from one language to another. Example: English to Hindi using a translation system.

Chatbots: Computer programs that communicate with users using natural language, such as customer-support or university-helpdesk bots.

Sentiment Analysis: Identifies the opinion or emotion in text, commonly positive, negative, or neutral.

Speech Recognition: Converts human speech into text. Examples include voice typing and voice assistants.

Text Classification: Assigns text to predefined categories, such as spam/not spam or sports/technology news.

Information Extraction: Extracts useful facts such as people, organizations, locations, dates, and other entities from text.

Search Engines: Processes search queries to understand the user's intent and retrieve relevant information.

Text Summarization: Creates a shorter version of a long document while keeping important information.

Question Answering: Answers questions asked in natural language.

Autocomplete / Predictive Text: Predicts likely words or phrases while a user is typing.

4. Challenges in NLP

Human language is complex, flexible, and context-dependent. This creates many challenges for computers.

Ambiguity: A word or sentence may have more than one meaning. Example: 'bank' can mean a financial institution or the side of a river.

Context Understanding: The meaning of a word can change depending on surrounding words. 'Apple' can refer to a fruit or a company.

Sarcasm: People may say something that appears positive but actually expresses criticism. Example: 'Great, my phone stopped working again.'

Idioms: Expressions cannot always be understood word-by-word. 'Raining cats and dogs' means raining heavily.

Synonyms: Different words can have similar meanings, such as big, large, and huge.

Multiple Languages: Different languages have different vocabularies, grammar, scripts, and sentence structures.

Spelling Errors: Informal or incorrect spelling such as 'goin' or 'clg' can make processing difficult.

Slang and Informal Language: Social media contains expressions such as LOL, BTW, gonna, and wanna.

Lack of Data: Some languages have limited training data, dictionaries, and annotated datasets.

Human Emotions: Emotion can be expressed indirectly and may require context to understand correctly.

Types of Ambiguity

Lexical ambiguity: One word has multiple meanings. Example: 'bank'.

Syntactic ambiguity: A sentence can have more than one grammatical interpretation. Example: 'I saw the boy with the telescope.'

Semantic ambiguity: The sentence structure may be clear but its meaning can still have multiple interpretations.

5. NLP Pipeline

The NLP pipeline is a sequence of steps used to convert raw human language into useful information.

Step	Purpose
1. Raw/Input Text	Receives text from a user, document, website, review, etc.
2. Text Preprocessing	Cleans and prepares the text.
3. Tokenization	Breaks text into smaller units called tokens.
4. Stop-word Removal	May remove common words that are not useful for a particular task.
5. Stemming / Lemmatization	Reduces words to root-like or dictionary base forms.
6. Text Representation	Converts text into numerical form such as vectors.
7. Feature Extraction	Identifies useful information/features.
8. NLP Model	Applies a machine learning or deep learning model.
9. Output	Produces the required prediction or result.

Pipeline to remember: Raw Text → Preprocessing → Tokenization → Stop-word Removal → Stemming/Lemmatization → Text Representation → Feature Extraction → NLP Model → Output

6. Text Preprocessing

Text preprocessing is the process of cleaning and transforming raw text into a suitable format for further NLP processing.

- Lowercasing
- Tokenization
- Punctuation removal
- Stop-word removal
- Stemming
- Lemmatization
- Handling special characters
- Removing unnecessary spaces
- Handling numbers, URLs, and other task-specific noise

7. Lowercasing

Lowercasing converts text into lowercase so that different capitalization forms are treated consistently.

Example: Natural Language Processing → natural language processing

It can reduce duplicate forms such as NLP, Nlp, and nlp. However, capitalization can carry useful information in some tasks, so it should not always be applied blindly.

8. Tokenization

Tokenization is the process of breaking text into smaller units called tokens. Tokens may be words, sentences, subwords, or characters.

Word tokenization: 'I love Natural Language Processing.' → I | love | Natural | Language | Processing

Sentence tokenization: A paragraph is divided into individual sentences.

Importance: Tokenization makes text easier for an NLP system to process because a large piece of text is divided into manageable units.

9. Stop-Word Removal

Stop words are commonly occurring words that may provide little useful information for some NLP tasks. Examples include the, is, am, are, a, an, of, in, on, and, and to.

Example: 'The student is studying in the classroom.' → 'student studying classroom'

Advantages

- Reduces the amount of text to process.
- Can reduce the number of features.
- May improve processing speed.

Important: Stop words are not always useless. Words such as 'not', 'no', and 'never' can change the meaning of a sentence. For example, removing 'not' from 'This movie is not good' can reverse the meaning.

10. Stemming

Stemming reduces words to a root-like form, usually by applying simple rules that remove prefixes or suffixes. The result may not be a valid English word.

Original	Possible stem
playing	play
played	play
plays	play
studies	studi
studying	studi
connected	connect

Advantages: simple, fast, reduces vocabulary size, and can help in search-related tasks.

Disadvantages: can create invalid words, may over-stem unrelated words to the same form, and may under-stem related words.

Common algorithms: Porter Stemmer, Snowball Stemmer, and Lancaster Stemmer.

11. Lemmatization

Lemmatization reduces a word to its correct dictionary or base form, called a **lemma**. It uses linguistic information and can consider vocabulary, morphology, grammar, and Part-of-Speech information.

Word	Lemma
running	run
ran	run
studies	study
better	good
mice	mouse
children	child

Lemmatization generally gives linguistically meaningful words, but it can require more processing than simple stemming.

12. Stemming vs Lemmatization

Feature	Stemming	Lemmatization
Method	Usually rule-based suffix/prefix removal	Uses linguistic information
Output	Root-like form	Dictionary/base form
Validity	May not be a valid word	Usually a valid word
Speed	Generally faster	Generally slower
Accuracy	Less linguistically accurate	Generally more accurate
Example	studies → studi	studies → study

> **Easy memory trick:** Stemming = cut the word. Lemmatization = understand the word and find its proper base form.

13. Complete Text Preprocessing Example

Original: 'The students are studying NLP!!! They are enjoying it.'

Lowercase: the students are studying nlp!!! they are enjoying it.

Tokenization: the | students | are | studying | nlp | they | are | enjoying | it

Punctuation removal: the | students | are | studying | nlp | they | are | enjoying | it

Stop-word removal: students | studying | nlp | enjoying

Stemming: student | studi | nlp | enjoy

Lemmatization: student | study | nlp | enjoy

14. Complete NLP Example

Input: 'I really loved this movie! It was amazing.'

Lowercase: i really loved this movie! it was amazing.

Tokenization: i | really | loved | this | movie | it | was | amazing

Stop-word removal: really | loved | movie | amazing

Lemmatization: really | love | movie | amazing

Text representation: The cleaned words are converted into numerical representations.

Model: A sentiment classification model processes the representation.

Output: Positive sentiment

15. Why Text Preprocessing is Important

- Raw text is often unstructured and noisy.
- It may contain unnecessary punctuation, repeated forms, spelling variations, and irrelevant words.
- Preprocessing makes text cleaner and more consistent.
- It can reduce the amount of data and vocabulary that a model needs to process.
- Good preprocessing can improve the usefulness of features and downstream NLP tasks.

16. Important Definitions for the Exam

Natural Language: Language used by humans for communication.

NLP: A field of AI that enables computers to process and work with human language.

Token: A basic unit produced during tokenization.

Tokenization: Breaking text into smaller units such as words or sentences.

Stop Words: Common words that may be removed for certain NLP tasks.

Stemming: Reducing a word to a root-like form using rules.

Lemmatization: Reducing a word to its proper dictionary/base form.

NLP Pipeline: A sequence of processing steps that converts raw language into useful output.

17. Important Exam Questions

1. What is Natural Language Processing? Explain its importance and applications.

2. Explain the NLP pipeline in detail with a suitable example.

3. What are the major challenges in Natural Language Processing?

4. Explain text preprocessing in NLP.

5. Explain tokenization and its types with examples.

6. What are stop words? Explain stop-word removal with examples.

7. Explain stemming with suitable examples.

8. Explain lemmatization with suitable examples.

9. Differentiate between stemming and lemmatization.

10. Explain the major applications of NLP.

18. Quick Revision Sheet

NLP: AI technology for processing human language.
Applications: Translation, chatbots, sentiment analysis, speech recognition, classification, information extraction, search, summarization, question answering, predictive text.
Challenges: Ambiguity, context, sarcasm, idioms, synonyms, multiple languages, spelling errors, slang, lack of data, emotions.
Pipeline: Raw Text → Preprocessing → Tokenization → Stop-word Removal → Stemming/Lemmatization → Text Representation → Feature Extraction → NLP Model → Output.
Stemming: Root-like form; may be invalid; usually faster.
Lemmatization: Dictionary/base form; linguistically meaningful; usually more processing.

19. High-Priority Topics for Exam Preparation

Topic	Priority
NLP definition, importance and applications	Very High
Challenges in NLP	Very High
NLP Pipeline	Very High
Text preprocessing	Very High
Tokenization	High
Stemming vs Lemmatization	Very High
Stop-word removal	High